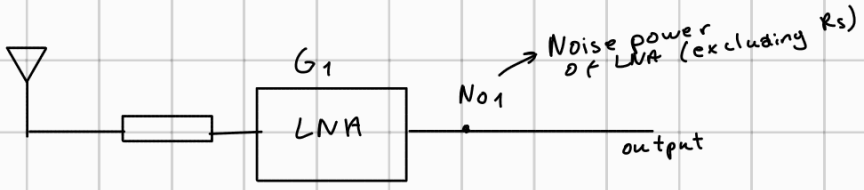


Input noise power $\frac{\overline{v_{in,n}^2}}{R_{in}} = \frac{kTR_s}{R_{in}}$ = kT in 1Hz BW

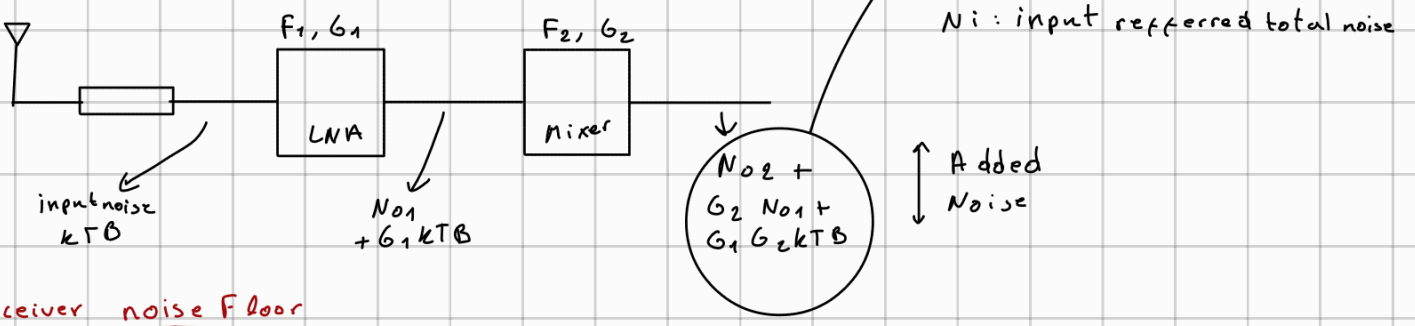
in B Bandwidth kTB



$$F = \frac{\text{Total noise power including } R_s}{A_v^2 \cdot \text{Noise from only } R_s} = \frac{\overline{v_{n,out}^2}}{A_v^2} \cdot \frac{1}{4kTR_s}$$

$$F_1 = \frac{N_{o1} + G_1 kTB}{G_1 \cdot kTB}$$ (girişteki gürültü)

$$N_{o1} = (F_1 - 1) G_1 kTB$$



Receiver noise Floor

$$F = \frac{N_o + GkTB}{G} \cdot \frac{1}{kTB}$$

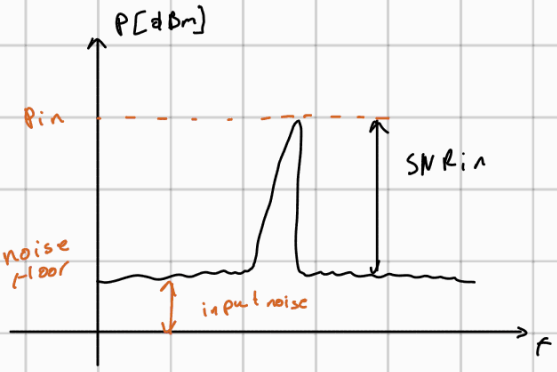
~~$N_o = (F-1) kTB \cdot G$~~

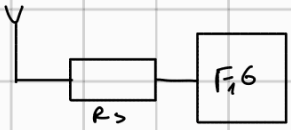
$$F\,kTB = \frac{N_o}{G} + kTB = \text{noise Floor}$$

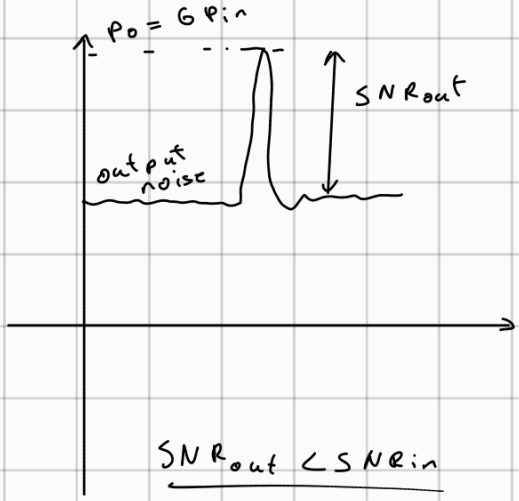
noise floor = $F\,kTB$ → temperature

F: Noise factor; k: Boltzman constant

Receiver SNR & Noise





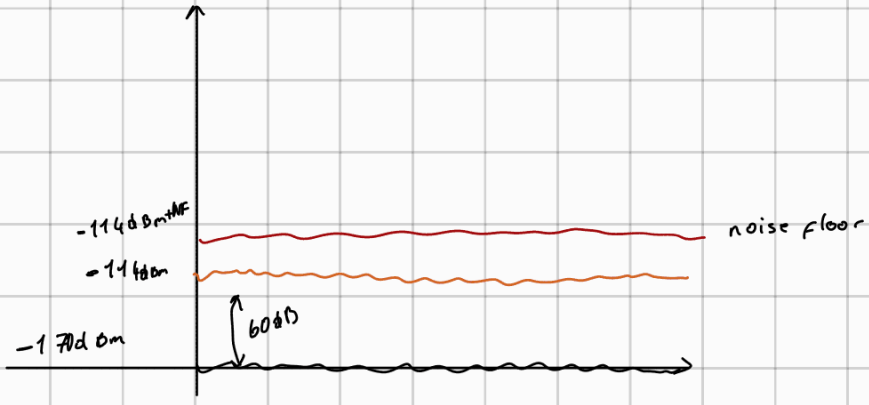


$SNR_{out} < SNR_{in}$

output noise = input Noise + NA

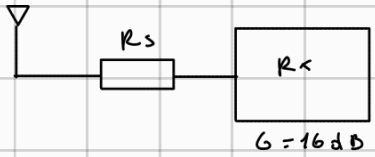
$$\text{noise floor} = FkTB = \underbrace{10\log F}_{NF} + \underbrace{10\log kT}_{-174\ \frac{dBm}{Hz}}{}^{\nearrow 25^\circ} + 10\log B$$

$$\text{noise floor} = -174\ \frac{dBm}{Hz} + 10\log \underbrace{B}_{10^6} + NF$$



Output referred total noise $= FkTB \cdot G = NF - 174\ \frac{dBm}{Hz} + 10\log B + 10\log G$

Ex:



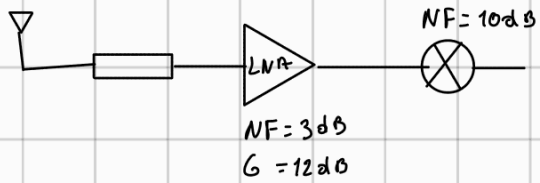
G = 16dB
NF = 10dB
B = 10 MHz

$$\text{noise floor} = FkTB = -174\ \frac{dBm}{Hz} + 10\log(10 \cdot 10^6) + \underbrace{10}_{NF}$$

$$= -174 + 70 + 10 = -94\ dBm$$

$$N_{out,t} = -94 + 16 = -78\ dBm$$

Ex:



Calculate NF total

$$F = F_1 + \frac{F_2 - 1}{G_1}$$ (Scalar olmalı)

$$= 10^{3/10} + \frac{10^{10/10} - 1}{10^{12/10}} = 2,57$$

$$NF = 10 \log F = 10 \log (2,57) = 4,1\,dB$$

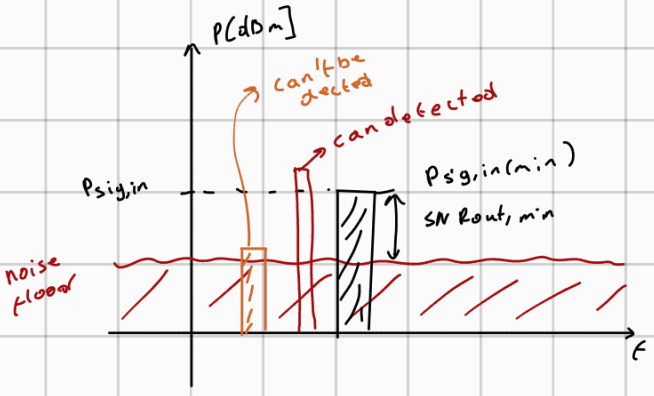
Calculate Noise floor ($B = 5MHz$)

$$\text{Noise floor} = -174\,\frac{dBm}{Hz} + 10 \log (5 \cdot 10^6) + 4.1$$

$$\text{Noise floor} = -102.9\,dBm$$

Sensitivity and Dynamic Range

Sensitivity is the minimum input signal which can be detected with the acceptable output SNR.



$$NF = \frac{(SNR)_{in}}{(SNR)_{out}} = \frac{\frac{P_{sig,in}}{P_{n,in}}}{SNR_{out}} = \frac{P_{sig,in}}{P_{n,in} \cdot SNR_{out}}$$

$$P_{sig,in} = \underbrace{P_{n,in}}_{kTB \text{ if } R_S = R_{in}} \cdot F \cdot SNR_{out}$$

$$P_{sig,in} = \underbrace{kTB \cdot F}_{\text{noise floor}} \cdot SNR_{out}$$

$$P_{sig,in}(min)\Big|_{dB} = \text{noise floor}_{dB} + SNR_{out_{dB}}$$

$$\underbrace{P_{sig,in(min)}\big|_{dB}}_{\text{sensitivity}} = -174\ \frac{dBm}{Hz} + 10\log B + NF + SNR_{out}\big|_{dB}$$

Ex: A GSM receiver has $SNR_{out(min)} = 12dB$

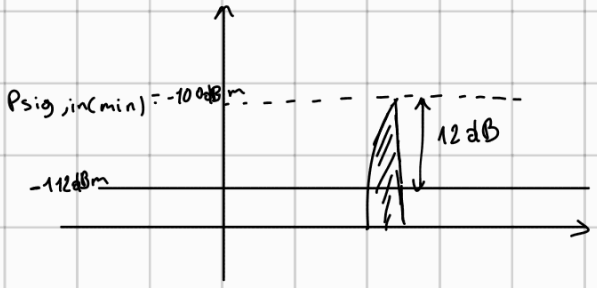
channel $BW = 20khz$

$NF = 9dB$

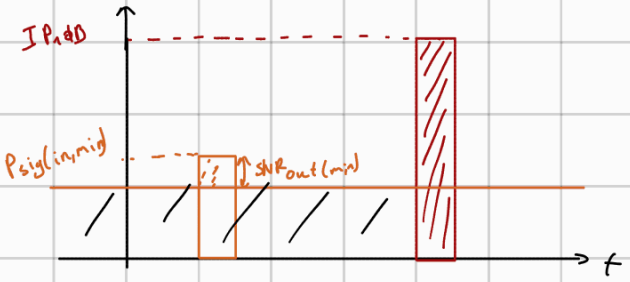
what is tthe minimum acceptable signal power?

$$P_{sig,in(min)} = -174\ \frac{dBm}{Hz} + \underbrace{10\log(20\cdot10^3)}_{\approx 43} + 9 + 12$$

$$= -112 + 12 = -100\ dBm$$



Dynamic Range

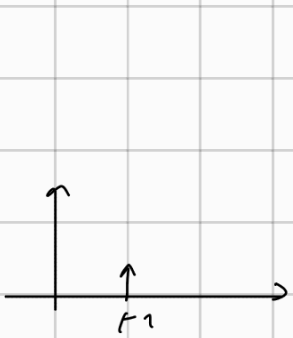


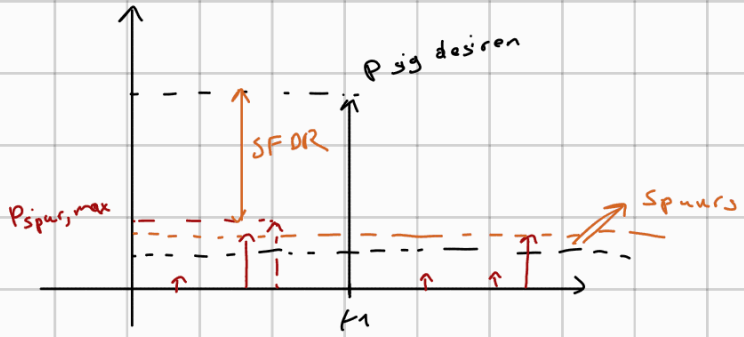
$$DR = IP_{1dB} - \underbrace{P_{sig,in(min)}\big|_{dB}}_{MDS} = IP_{1dB} - MDS$$

MDS: Minimum detectable signal

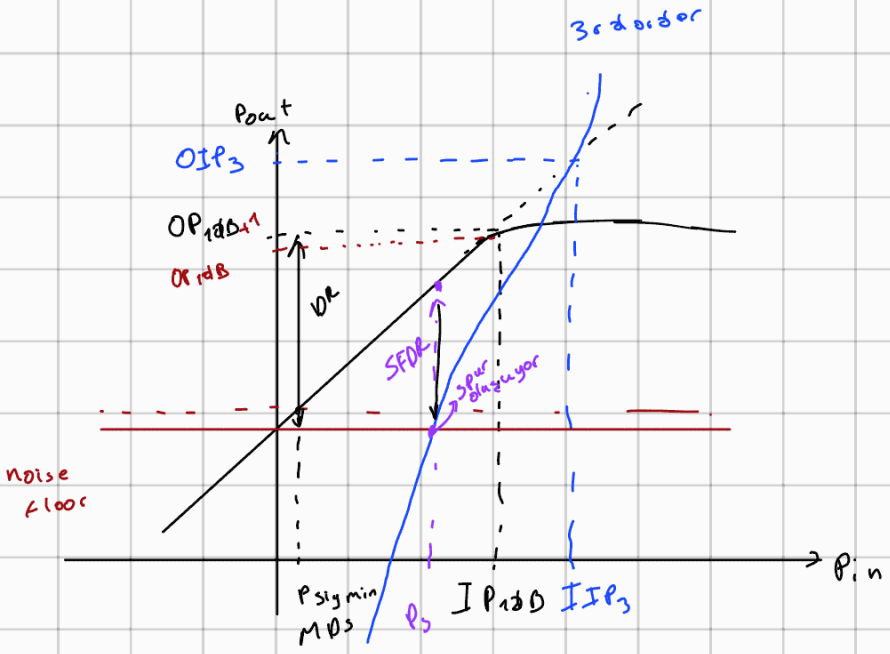
SFDR

Spur-free Dynamic Range





$$SFDR = P_{sig,desired} - P_{spur(max)}$$





$$SFDR = \frac{2\,(P_{IIP3} + 174 - NF - 10\log B)}{3} - SN\,R_{out(min)}$$